

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-
scale=1.0">

<title>PulseInvest — Smart DCA Advisor</title>

<link
href="https://fonts.googleapis.com/css2?family=DM+Serif+Display:ital@0;1&family=
DM+Sans:opsz,wght@9..40,300;9..40,400;9..40,500;9..40,600&family=DM+Mono:wg
ht@400;500&display=swap" rel="stylesheet">

<style>

:root{

  --bg:#f7f5f0;--bg2:#eeeeae2;--surface:#fff;--border:#e0dbd0;

  --ink:#1a1814;--ink2:#6b6560;--ink3:#aba69e;

  --accent:#c8813a;--al:#f5ead8;

  --green:#2d7a4f;--gl:#d6f0e0;

  --red:#b83232;--rl:#fde8e6;

  --r:14px;--sh:0 2px 16px rgba(0,0,0,0.06);

  /* Rating colours */

  --r1:#1a6b3a;--r1l:#c8f0d8; /* Strong Buy */

  --r2:#2d7a4f;--r2l:#d6f0e0; /* Buy */

  --r3:#b07d2a;--r3l:#fdf0d0; /* Neutral */

  --r4:#b83232;--r4l:#fde8e6; /* Reduce */

  --r5:#7a1a1a;--r5l:#fcd0d0; /* Strong Reduce */
```

```
}
```

```
*{box-sizing:border-box;margin:0;padding:0;}
```

```
body{background:var(--bg);color:var(--ink);font-family:'DM Sans',sans-serif;min-height:100vh;font-size:15px;-webkit-font-smoothing:antialiased;}
```

```
nav{display:flex;justify-content:space-between;align-items:center;padding:16px 20px;background:var(--surface);border-bottom:1px solid var(--border);position:sticky;top:0;z-index:100;box-shadow:0 1px 8px rgba(0,0,0,.05);}
```

```
.logo{font-family:'DM Serif Display',serif;font-size:20px;letter-spacing:-.3px;}
```

```
.logo span{color:var(--accent);}
```

```
.lang-toggle{display:flex;background:var(--bg2);border-radius:20px;padding:3px;gap:2px;}
```

```
.lang-btn{padding:4px 14px;border-radius:16px;border:none;background:transparent;font-size:12px;font-weight:500;color:var(--ink2);cursor:pointer;font-family:'DM Sans',sans-serif;transition:all .2s;}
```

```
.lang-btn.active{background:var(--surface);color:var(--ink);box-shadow:0 1px 4px rgba(0,0,0,.1);}
```

```
.wrap{max-width:520px;margin:0 auto;padding:24px 16px 80px;}
```

```
.hero{text-align:center;padding:28px 0 24px;}
```

```
.hero-tag{display:inline-block;font-size:10px;letter-spacing:2.5px;text-transform:uppercase;color:var(--accent);font-weight:600;margin-bottom:12px;font-family:'DM Mono',monospace;}
```

```
.hero h1{font-family:'DM Serif Display',serif;font-size:30px;line-height:1.18;letter-spacing:-.5px;margin-bottom:10px;}
```

```
.hero h1 em{font-style:italic;color:var(--accent);}
```

```
.hero p{color:var(--ink2);font-size:13.5px;line-height:1.65;max-width:360px;margin:0 auto 14px;}
```

```

.disc{display:inline-block;background:var(--bg2);border:1px solid var(--
border);border-radius:20px;padding:5px 14px;font-size:11px;color:var(--ink3);font-
family:'DM Mono',monospace;}

.tab-bar{display:flex;background:var(--bg2);border-radius:12px;padding:4px;margin-
bottom:20px;gap:3px;}

.tab-btn{flex:1;padding:9px 6px;border:none;background:transparent;border-
radius:9px;font-size:12px;font-weight:500;color:var(--
ink2);cursor:pointer;transition:all .2s;font-family:'DM Sans',sans-serif;}

.tab-btn.active{background:var(--surface);color:var(--ink);box-shadow:0 1px 6px
rgba(0,0,0,.08);}

.card{background:var(--surface);border:1px solid var(--border);border-radius:var(--
r);padding:20px;margin-bottom:14px;box-shadow:var(--sh);}

.card-label{font-size:10px;letter-spacing:2px;text-transform:uppercase;color:var(--
ink3);font-family:'DM Mono',monospace;margin-bottom:14px;font-weight:500;}

.field{margin-bottom:13px;}.field:last-child{margin-bottom:0;}

label{display:block;font-size:12px;font-weight:600;color:var(--ink2);margin-
bottom:5px;}

.lsub{font-weight:400;color:var(--ink3);font-size:11px;margin-left:4px;}

input[type=text],input[type=number],select{width:100%;background:var(--
bg);border:1.5px solid var(--border);border-radius:10px;color:var(--ink);font-
size:15px;font-family:'DM Mono',monospace;padding:11px
14px;outline:none;transition:border-color .2s,box-shadow .2s;-webkit-
appearance:none;}

input:focus,select:focus{border-color:var(--accent);box-shadow:0 0 0 3px
rgba(200,129,58,.12);background:var(--surface);}

input::placeholder{color:var(--ink3);font-size:13px;font-family:'DM Sans',sans-serif;}

select{cursor:pointer;}

.g2{display:grid;grid-template-columns:1fr 1fr;gap:12px;}.g2 .field{margin-bottom:0;}

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```

.macro-opts{display:flex;gap:8px;margin-top:4px;}

.mopt{flex:1;padding:10px 6px;border:1.5px solid var(--border);border-
radius:10px;background:var(--bg);text-
align:center;cursor:pointer;transition:all .2s;font-size:12px;font-
weight:500;color:var(--ink2);}

.mopt:hover{border-color:var(--accent);}

.mopt.sel{border-color:var(--accent);background:var(--al);color:var(--ink);}

.mopt-emoji{display:block;font-size:18px;margin-bottom:4px;}

.macro-note{font-size:11px;color:var(--ink3);margin-top:10px;line-height:1.55;}

.calc-btn{width:100%;padding:16px;background:var(--ink);border:none;border-
radius:12px;color:#fff;font-size:15px;font-weight:600;font-family:'DM Sans',sans-
serif;cursor:pointer;letter-spacing:.3px;transition:all .2s;margin-top:4px;}

.calc-btn:hover{background:#2d2a26;transform:translateY(-1px);box-shadow:0 6px
20px rgba(0,0,0,.15);}

.calc-btn:active{transform:translateY(0);}

.calc-btn:disabled{opacity:.6;cursor:not-allowed;transform:none;box-shadow:none;}

/* —— RATING BADGE —— */

.rating-banner{border-radius:12px;padding:16px 20px;margin-
bottom:14px;display:flex;align-items:center;gap:14px;}

.rating-dot{width:48px;height:48px;border-radius:50%;display:flex;align-
items:center;justify-content:center;font-size:22px;flex-shrink:0;}

.rating-text-wrap{flex:1;}

.rating-label{font-size:11px;letter-spacing:2px;text-transform:uppercase;font-
family:'DM Mono',monospace;font-weight:600;margin-bottom:3px;}

.rating-name{font-size:18px;font-weight:700;font-family:'DM Serif Display',serif;}

.rating-mult{font-size:13px;margin-top:2px;font-family:'DM Mono',monospace;font-

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weight:500;}

/* colours per tier */

.tier1{background:var(--r1l);border:1.5px solid var(--r1);}.tier1 .rating-
label,.tier1 .rating-name,.tier1 .rating-mult{color:var(--r1);}

.tier2{background:var(--r2l);border:1.5px solid var(--r2);}.tier2 .rating-
label,.tier2 .rating-name,.tier2 .rating-mult{color:var(--r2);}

.tier3{background:var(--r3l);border:1.5px solid var(--r3);}.tier3 .rating-
label,.tier3 .rating-name,.tier3 .rating-mult{color:var(--r3);}

.tier4{background:var(--r4l);border:1.5px solid var(--r4);}.tier4 .rating-
label,.tier4 .rating-name,.tier4 .rating-mult{color:var(--r4);}

.tier5{background:var(--r5l);border:1.5px solid var(--r5);}.tier5 .rating-
label,.tier5 .rating-name,.tier5 .rating-mult{color:var(--r5);}


/* —— RESULT HERO —— */

#result{display:none;}#result.show{display:block;}

.r-hero{background:var(--ink);border-radius:var(--r);padding:28px 24px;text-
align:center;margin-bottom:14px;position:relative;overflow:hidden;}

.r-hero::before{content:'';position:absolute;top:-40px;right:-
40px;width:140px;height:140px;background:radial-
gradient(circle,rgba(200,129,58,.2),transparent 70%);border-radius:50%;}

.r-eyebrow{font-size:10px;letter-spacing:2.5px;text-
transform:uppercase;color:rgba(255,255,255,.4);font-family:'DM
Mono',monospace;margin-bottom:10px;}

.r-amount{font-family:'DM Serif Display',serif;font-size:52px;color:#fff;line-
height:1;margin-bottom:6px;}

.r-sub{font-size:13px;color:rgba(255,255,255,.45);margin-bottom:16px;}

.r-badge{display:inline-block;padding:6px 18px;border-radius:20px;font-

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size:13px;font-weight:600;font-family:'DM Mono',monospace;}

.r-badge.up{background:rgba(45,122,79,.3);color:#5dbc87;}

.r-badge.dn{background:rgba(184,50,50,.3);color:#e87060;}

.r-badge.nt{background:rgba(255,255,255,.1);color:rgba(255,255,255,.55);}

.macro-impact{margin-top:12px;font-size:12px;font-family:'DM
Mono',monospace;padding:8px 14px;border-radius:8px;display:inline-block;}

.macro-impact.pos{background:rgba(45,122,79,.2);color:#5dbc87;}

.macro-impact.neg{background:rgba(184,50,50,.2);color:#e87060;}

.macro-impact.zero{background:rgba(255,255,255,.08);color:rgba(255,255,255,.4);}

/* —— CHIPS —— */

.chips{display:flex;flex-wrap:wrap;gap:8px;margin-bottom:14px;}

.chip{display:flex;flex-direction:column;align-items:center;background:var(--
bg2);border:1px solid var(--border);border-radius:10px;padding:10px
12px;flex:1;min-width:76px;text-align:center;}

.chip-label{font-size:9px;letter-spacing:1.5px;text-transform:uppercase;color:var(--
ink3);font-family:'DM Mono',monospace;margin-bottom:5px;}

.chip-val{font-size:17px;font-weight:600;font-family:'DM
Mono',monospace;color:var(--ink);}

.chip-sub{font-size:10px;margin-top:3px;font-weight:500;color:var(--ink3);}

.chip.g{border-color:#a8d8bb;background:var(--gl);}.chip.g .chip-val{color:var(--
green);}

.chip.r{border-color:#f0b0a8;background:var(--rl);}.chip.r .chip-val{color:var(--red);}

.chip.o{border-color:#e8c898;background:var(--al);}.chip.o .chip-val{color:var(--
accent);}

```

```
/* —— SCORE BARS —— */
```

```
.srow{display:flex;align-items:center;gap:10px;margin-bottom:9px;}.srow:last-child{margin-bottom:0;}
```

```
.sname{font-size:12px;color:var(--ink2);width:110px;flex-shrink:0;font-weight:500;}
```

```
.sbar-bg{flex:1;height:6px;background:var(--bg2);border-radius:3px;overflow:hidden;}
```

```
.sbar-fill{height:100%;border-radius:3px;transition:width .8s ease;}
```

```
.snum{font-size:12px;font-family:'DM Mono',monospace;color:var(--ink2);width:28px;text-align:right;flex-shrink:0;}
```

```
/* —— REASON —— */
```

```
.reason-list{list-style:none;}
```

```
.reason-list li{display:flex;gap:10px;font-size:13px;line-height:1.65;color:var(--ink2);padding:9px 0;border-bottom:1px solid var(--border);}
```

```
.reason-list li:last-child{border-bottom:none;}
```

```
.ri{flex-shrink:0;font-size:14px;margin-top:1px;}
```

```
/* —— COMPARE —— */
```

```
.cmp-grid{display:grid;grid-template-columns:1fr 1fr;gap:10px;margin-bottom:14px;}
```

```
.cmp-box{background:var(--surface);border:1.5px solid var(--border);border-radius:12px;padding:16px;text-align:center;}
```

```
.cmp-box.win{border-color:#a8d8bb;background:var(--gl);}
```

```
.cmp-label{font-size:10px;letter-spacing:1.5px;text-transform:uppercase;color:var(--ink3);font-family:'DM Mono',monospace;margin-bottom:8px;}
```

```
.cmp-val{font-size:20px;font-weight:700;font-family:'DM Mono',monospace;color:var(--ink);}
```

```

.cmp-box.win .cmp-val{color:var(--green);}

.cmp-sub{font-size:11px;color:var(--ink3);margin-top:4px;}

.cmp-diff{font-size:12px;font-weight:600;color:var(--green);margin-top:4px;}

/* —— CHART —— */

.chart-card{margin-bottom:14px;}

.chart-title{font-size:13px;font-weight:600;color:var(--ink);margin-bottom:14px;}

canvas{width:100%!important;}

/* —— ACTION BUTTONS (bottom of result) —— */

.action-row{display:grid;grid-template-columns:1fr 1fr;gap:10px;margin-bottom:14px;}

.act-btn{padding:14px 10px;border-radius:12px;font-size:13px;font-weight:500;font-family:'DM Sans',sans-serif;cursor:pointer;transition:all .2s;text-align:center;border:1.5px solid var(--border);background:transparent;color:var(--ink2);}

.act-btn:hover{border-color:var(--accent);color:var(--accent);}

.act-btn.primary{background:var(--ink);color:#fff;border-color:var(--ink);}

.act-btn.primary:hover{background:#2d2a26;border-color:#2d2a26;}

.act-btn.saved{background:var(--gl);border-color:#a8d8bb;color:var(--green);}

/* —— WATCHLIST —— */

#wlPanel{display:none;}#wlPanel.show{display:block;}

.wl-empty{text-align:center;padding:40px 20px;color:var(--ink3);font-size:13px;}

.wl-card{background:var(--surface);border:1px solid var(--border);border-

```



```

radius:12px;padding:16px;margin-bottom:10px;display:flex;align-
items:center;gap:12px;box-shadow:var(--sh);}

.wl-ticker{font-family:'DM Serif Display',serif;font-size:22px;color:var(--
ink);width:70px;flex-shrink:0;}

.wl-info{flex:1;}

.wl-budget{font-size:11px;color:var(--ink3);font-family:'DM Mono',monospace;}

.wl-amount{font-size:20px;font-weight:700;font-family:'DM
Mono',monospace;margin-top:2px;}

.wl-badge{display:inline-block;padding:3px 10px;border-radius:20px;font-
size:11px;font-weight:600;font-family:'DM Mono',monospace;margin-top:4px;}

.wl-remove{background:none;border:none;color:var(--ink3);font-
size:18px;cursor:pointer;padding:4px;flex-shrink:0;}

.wl-remove:hover{color:var(--red);}

.wl-loading{font-size:11px;color:var(--ink3);font-family:'DM
Mono',monospace;margin-top:4px;}

.refresh-btn{width:100%;padding:12px;background:transparent;border:1.5px solid
var(--border);border-radius:12px;color:var(--ink2);font-size:13px;font-family:'DM
Sans',sans-serif;cursor:pointer;transition:all .2s;margin-bottom:14px;}

.refresh-btn:hover{border-color:var(--accent);color:var(--accent);}

/* —— BACKTEST —— */

#btPanel{display:none;}#btPanel.show{display:block;}

.bt-row{display:flex;justify-content:space-between;padding:10px 0;border-
bottom:1px solid var(--border);font-size:13px;}

.bt-row:last-child{border-bottom:none;}

.bt-key{color:var(--ink2);}

.bt-val{font-family:'DM Mono',monospace;font-weight:600;color:var(--ink);}

```

```

.bt-val.g{color:var(--green);}

/* —— MISC —— */

.warn-box{background:#fff8f0;border:1px solid #e8c898;border-
radius:10px;padding:12px 14px;font-size:12px;color:#8a5a20;margin-top:10px;line-
height:1.6;display:none;}

.warn-box.show{display:block;}

.err-box{background:var(--rl);border:1px solid #f0b0a8;border-
radius:10px;padding:12px 14px;font-size:12px;color:var(--red);margin-top:10px;line-
height:1.6;display:none;}

.err-box.show{display:block;}

.copy-btn{width:100%;padding:14px;background:transparent;border:1.5px solid var(-
-border);border-radius:12px;color:var(--ink2);font-size:13px;font-weight:500;font-
family:'DM Sans',sans-serif;cursor:pointer;transition:all .2s;}

.copy-btn:hover{border-color:var(--accent);color:var(--accent);}

.spin{display:inline-block;width:14px;height:14px;border:2px solid
rgba(255,255,255,.3);border-top-color:#fff;border-radius:50%;animation:sp .7s linear
infinite;vertical-align:middle;margin-right:6px;}

.spin-dk{border-color:rgba(0,0,0,.15);border-top-color:var(--ink);}

@keyframes sp{to{transform:rotate(360deg);}}

@keyframes
fadeUp{from{opacity:0;transform:translateY(12px);}to{opacity:1;transform:translateY
(0);}}

.fu{animation:fadeUp .4s ease forwards;}

@media(max-width:400px){.r-amount{font-size:42px;}.chip{padding:8px 8px;}.chip-
val{font-size:15px;}.action-row{grid-template-columns:1fr;}}

</style>

```

```

</head>

<body>

<nav>

  <div class="logo">Pulse<span>Invest</span></div>

  <div class="lang-toggle">

    <button class="lang-btn active" onclick="setLang('en')"
id="btnEn">EN</button>

    <button class="lang-btn" onclick="setLang('zh')" id="btnZh">中文</button>

  </div>

</nav>

<div class="wrap">

  <div class="hero">

    <div class="hero-tag" id="t-tag">Smart DCA Advisor</div>

    <h1 id="t-h1">Invest <em>smarter,</em><br>not just regularly.</h1>

    <p id="t-sub">Stop buying the same amount every month. Let the data tell you
when to go bigger — and when to ease back.</p>

    <div class="disc" id="t-disc">Not investment advice &nbsp;·&nbsp; Past ≠
Future</div>

  </div>

  <div class="tab-bar">

    <button class="tab-btn active" onclick="switchTab('advisor')" id="tabA">📊
<span id="t-tab1">This Month</span></button>

    <button class="tab-btn" onclick="switchTab('watchlist')" id="tabW">★ <span
id="t-tab3">Watchlist</span></button>

```

```
<button class="tab-btn" onclick="switchTab('backtest')" id="tabB"><img alt="clock icon" data-bbox="738 91 765 108"/> <span id="t-tab2">Backtest</span></button>
```

```
</div>
```

```
<!-- ===== ADVISOR ===== -->
```

```
<div id="aPanel">
```

```
<div class="card">
```

```
<div class="card-label" id="t-s1">Stock</div>
```

```
<div class="field">
```

```
<label id="t-tklabel">Ticker <span class="lsub" id="t-tkhint">US stocks · e.g.  
AAPL · SPY · NVDA · MSFT</span></label>
```

```
<input type="text" id="ticker" placeholder="AAPL"  
autocapitalize="characters" autocorrect="off" autocomplete="off"  
oninput="this.value=this.value.toUpperCase()">
```

```
<div class="err-box" id="errBox"></div>
```

```
<div class="warn-box" id="warnBox"></div>
```

```
</div>
```

```
</div>
```

```
<div class="card">
```

```
<div class="card-label" id="t-s2">Your Plan</div>
```

```
<div class="field">
```

```
<label id="t-blable">Budget per Period (USD)</label>
```

```
<input type="number" id="budget" placeholder="500" min="10"  
step="10">
```

```
</div>
```

```
<div class="g2">
```

```
    <div class="field"><label id="t-plabel">Frequency</label>
```

```
        <select id="period"><option value="weekly" id="t-  
pw">Weekly</option><option value="biweekly" id="t-pb">Bi-  
weekly</option><option value="monthly" selected id="t-  
pm">Monthly</option></select>
```

```
    </div>
```

```
    <div class="field"><label id="t-hlabel">Time Horizon</label>
```

```
        <select id="horizon"><option value="12">1 Year</option><option  
value="24">2 Years</option><option value="36" selected>3 Years</option><option  
value="60">5 Years</option><option value="120">10 Years</option></select>
```

```
    </div>
```

```
</div>
```

```
    <div class="field"><label id="t-elabel">Already Running For</label>
```

```
        <select id="executed"><option value="0" id="t-e0">Just  
starting</option><option value="3">3 months</option><option value="6">6  
months</option><option value="12">1 year</option><option value="24">2  
years</option><option value="36">3+ years</option></select>
```

```
    </div>
```

```
</div>
```

```
<div class="card">
```

```
    <div class="card-label" id="t-s3">Market Sentiment <span class="lsub" id="t-  
s3sub">your gut feel · ±5% weight</span></div>
```

```
    <div class="macro-opts">
```

```
        <div class="mopt sel" data-val="1" onclick="selMacro(this)"><span  
class="mopt-emoji"> </span><span id="t-m1">Bearish</span></div>
```

```
        <div class="mopt" data-val="0" onclick="selMacro(this)"><span
```

class="mopt-emoji">😐Neutral</div>

<div class="mopt" data-val="-1" onclick="selMacro(this)"><span
class="mopt-emoji">📈Bullish</div>

</div>

<p class="macro-note" id="t-mnote">Your read on the overall market — Fed
news, headlines, how things feel. Adds or subtracts up to 5%. Small but
transparent.</p>

</div>

<button class="calc-btn" id="calcBtn" onclick="calculate()"><span id="t-
cbtn">Calculate This Month's Amount →</button>

<!-- RESULT -->

<div id="result">

<!-- Rating Banner -->

<div id="ratingBanner" class="rating-banner fu">

<div class="rating-dot" id="ratingDot"></div>

<div class="rating-text-wrap">

<div class="rating-label" id="t-ratinglabel">SIGNAL RATING</div>

<div class="rating-name" id="ratingName">—</div>

<div class="rating-mult" id="ratingMult">—</div>

</div>

</div>

<!-- Market Snapshot chips -->

<div class="card fu">

<div class="card-label" id="t-mklab">Market Snapshot</div>

<div class="chips">

<div class="chip" id="cPrice"><div class="chip-label">PRICE</div><div class="chip-val" id="cvPrice">—</div><div class="chip-sub" id="csPrice"></div></div>

<div class="chip" id="cRSI"><div class="chip-label">RSI 14</div><div class="chip-val" id="cvRSI">—</div><div class="chip-sub" id="csRSI"></div></div>

<div class="chip" id="cDD"><div class="chip-label" id="t-offhigh">OFF HIGH</div><div class="chip-val" id="cvDD">—</div><div class="chip-sub" id="csDD"></div></div>

<div class="chip" id="cTrend"><div class="chip-label" id="t-trendlab">TREND</div><div class="chip-val" id="cvTrend">—</div><div class="chip-sub" id="csTrend"></div></div>

</div>

</div>

<!-- Amount -->

<div class="r-hero fu">

<div class="r-eyebrow" id="t-rlabel">Suggested Investment</div>

<div class="r-amount" id="rAmt">—</div>

<div class="r-sub" id="rSub"></div>

<div class="r-badge nt" id="rBadge">×1.0</div>

<div class="macro-impact zero" id="macroImpact"></div>

</div>

<!-- Signal breakdown -->

```

<div class="card fu">

    <div class="card-label" id="t-siglab">Signal Breakdown</div>

    <div class="srow"><div class="sname">RSI <span style="color:var(--ink3);font-size:10px;">(45%)</span></div><div class="sbar-bg"><div class="sbar-fill" id="sRSI"></div></div><div class="snum" id="snRSI">—</div></div>

    <div class="srow"><div class="sname" id="t-sdd">Off High <span style="color:var(--ink3);font-size:10px;">(35%)</span></div><div class="sbar-bg"><div class="sbar-fill" id="sDD"></div></div><div class="snum" id="snDD">—</div></div>

    <div class="srow"><div class="sname" id="t-str">Trend <span style="color:var(--ink3);font-size:10px;">(20%)</span></div><div class="sbar-bg"><div class="sbar-fill" id="sTrend"></div></div><div class="snum" id="snTrend">—</div></div>

</div>

```

<!-- Reasons -->

```

<div class="card fu">

    <div class="card-label" id="t-rlab">Why This Amount</div>

    <ul class="reason-list" id="reasonList"></ul>

</div>

```

<!-- Compare -->

```

<div class="cmp-grid fu">

    <div class="cmp-box"><div class="cmp-label" id="t-fixed">Fixed DCA</div><div class="cmp-val" id="cFixed">—</div><div class="cmp-sub" id="t-shares">Shares accumulated</div></div>

    <div class="cmp-box win"><div class="cmp-label" id="t-smart">Smart

```


DCA</div><div class="cmp-val" id="cSmart">—</div><div class="cmp-sub" id="t-shares2">Shares accumulated</div><div class="cmp-diff" id="cDiff"></div></div>

</div>

<p style="font-size:11px;color:var(--ink3);text-align:center;margin-top:-8px;margin-bottom:14px;" id="t-cmpnote">Simulated over your selected time horizon using current price</p>

<!-- Chart -->

<div class="card chart-card fu">

<div class="chart-title" id="t-charttitle">Smart vs Fixed DCA — Share Accumulation</div>

<canvas id="growthChart" height="200"></canvas>

</div>

<!-- Action buttons -->

<div class="action-row fu">

<button class="act-btn primary" onclick="resetAdvisor()" id="analyzeOtherBtn"> Analyze Another</button>

<button class="act-btn" id="saveBtn" onclick="toggleSave()"> Save to Watchlist</button>

</div>

<button class="copy-btn" onclick="copyResult()" id="copyBtn"> Copy This Month's Summary</button>

</div>

</div>

```
<!-- ===== WATCHLIST ===== -->
```

```
<div id="wlPanel">
```

```
    <button class="refresh-btn" onclick="refreshWatchlist()" id="refreshBtn">
<span id="t-refresh">Refresh All</span></button>
```

```
    <div id="wlContent"></div>
```

```
</div>
```

```
<!-- ===== BACKTEST ===== -->
```

```
<div id="btPanel">
```

```
    <div class="card">
```

```
        <div class="card-label" id="t-bts1">Backtest Setup</div>
```

```
        <div class="field"><label id="t-bttklab">Ticker Symbol</label><input
type="text" id="btTicker" placeholder="SPY" autocapitalize="characters"
autocorrect="off" oninput="this.value=this.value.toUpperCase()"></div>
```

```
        <div class="g2">
```

```
            <div class="field"><label id="t-btstlab">Start (YYYY-MM)</label><input
type="text" id="btStart" placeholder="2019-01"></div>
```

```
            <div class="field"><label id="t-btenlab">End (YYYY-MM)</label><input
type="text" id="btEnd" placeholder="2024-12"></div>
```

```
        </div>
```

```
        <div class="field"><label id="t-btblab">Monthly Amount (USD)</label><input
type="number" id="btBudget" placeholder="500" min="10" step="10"></div>
```

```
    </div>
```

```
    <button class="calc-btn" onclick="runBacktest()" id="btBtn"><span id="t-
btbtn">Run Backtest →</span></button>
```

```
    <div id="btResult" style="display:none">
```


,bts1:'回測設定',bttklab:'股票代號',btstlab:'開始日期（YYYY-MM）',btenlab:'結束日期（YYYY-MM）',btblab:'每月金額（美元）',btbtn:'開始回測 →',btrlab:'回測結果',btchart:'歷史累積股數',btnote:'使用實際歷史月收盤價，智能定投倍數依當時指標回溯套用。',

tiers:['強力買進','建議買進','中性','建議減碼','強力減碼'],

tierEmoji:['🚀','✅','➡️','⚠️','🛑']

};

let lang='en';

function setLang(l){

lang=l;

document.getElementById('btnEn').classList.toggle('active',l==='en');

document.getElementById('btnZh').classList.toggle('active',l==='zh');

const t=T[l];

['tag','h1','sub','disc','tab1','tab2','tab3','s1','tklabel','tkhint','s2','blabel','plabel','hlabe
l','elabel','pw','pb','pm','e0','s3','s3sub','m1','m2','m3','mnote','cbtn','ratinglabel','mkl
ab','offhigh','trendlab','rlabel','siglab','sdd','str','rlab','fixed','smart','shares','shares2','
cmpnote','chartitle','other','save','cpbtn','refresh','bts1','bttklab','btstlab','btenlab','bt
blab','btbtn','btrlab','btchart','btnote'].forEach(k=>{const
el=document.getElementById('t-'+k);if(el&&t[k]!==undefined)el.innerHTML=t[k];});

document.documentElement.lang=l==='zh'?'zh-TW':'en';

}

function switchTab(t){

['aPanel','wlPanel','btPanel'].forEach(id=>{

const el=document.getElementById(id);

if(el)el.style.display='none';

```

        el&&el.classList&&el.classList.remove('show');

    });

['tabA','tabW','tabB'].forEach(id=>document.getElementById(id).classList.remove('active'));

if(t==='advisor'){document.getElementById('aPanel').style.display='block';document.getElementById('tabA').classList.add('active');}

    else
if(t==='watchlist'){document.getElementById('wlPanel').style.display='block';document.getElementById('tabW').classList.add('active');renderWatchlist();}

else{document.getElementById('btPanel').style.display='block';document.getElementById('tabB').classList.add('active');}

}

let macroVal=1;

function
selMacro(el){document.querySelectorAll('.mopt').forEach(e=>e.classList.remove('sel'));el.classList.add('sel');macroVal=parseInt(el.getAttribute('data-val'));}

// —— MATH ——

function calcRSI(closes,period){

    const n=Math.min(period,closes.length-1);let g=0,l=0;

    for(let i=closes.length-n;i<closes.length;i++){const d=closes[i]-closes[i-1];if(d>0)g+=d;else l-=d;}

    const ag=g/n,al=l/n;if(al===0)return 100;return 100-(100/(1+ag/al));

```

```

}

function avg(arr){return arr.reduce((a,b)=>a+b,0)/arr.length;}

function calcTrend(price,ma50,ma200){let s=50;if(price>ma50)s+=20;else s-=20;if(price>ma200)s+=20;else s-=20;if(ma50>ma200)s+=10;else s-=10;return Math.max(0,Math.min(100,s));}

function calcMult(rsi,dd,trend){

    const rs=100-rsi,ds=Math.min(100,Math.max(0,-dd*2.5+50)),comp=rs*.45+ds*.35+trend*.20;

    // Max mult now 1.95

    return Math.max(.4,Math.min(1.95,Math.round((.4+(comp/100)*1.55)*10)/10));

}


// —— RATING TIER ——

// Tier 1: mult >= 1.6 | Tier 2: 1.2-1.6 | Tier 3: 0.9-1.2 | Tier 4: 0.6-0.9 | Tier 5: < 0.6

function getTier(mult){

    if(mult>=1.6)return 0;

    if(mult>=1.2)return 1;

    if(mult>=0.9)return 2;

    if(mult>=0.6)return 3;

    return 4;

}

const tierClass=['tier1','tier2','tier3','tier4','tier5'];

const tierDotBg=['#1a6b3a','#2d7a4f','#b07d2a','#b83232','#7a1a1a'];

// —— 1. 換一個全新的函式名字，並加入時間戳記強制破壞快取 ——

```

// —— 全新防封鎖版本 (避開手機阻擋器攔截) ——

```
async function getFreshTwelveDataPrice(ticker){
```

```
    const zh = lang === 'zh';
```

```
    const t = ticker.toUpperCase();
```

// 💡 改用公共、對前端瀏覽器 100% 友善、不會被手機當成廣告阻擋的公共日 K 線節點

```
    const url =
```

```
    `https://allorigins.win{encodeURIComponent(`https://yahoo.com{t}?interval=1d&range=1y`)}&_=${Date.now()}`;
```

```
    try {
```

```
        const res = await fetch(url);
```

```
        if (!res.ok) throw new Error("Server communication failed");
```

```
        const wrapper = await res.json();
```

```
        if (!wrapper.contents) throw new Error("No contents retrieved");
```

```
        const json = JSON.parse(wrapper.contents);
```

```
        if (!json.chart || !json.chart.result || !json.chart.result[0]) {
```

```
            throw new Error("Invalid stock ticker data structure");
```

```
        }
```

```
        const r = json.chart.result[0];
```

```
        const rawCloses = r.indicators.quote[0].close;
```



```
// 過濾無效數據

const closes = rawCloses.filter(v => v !== null && v > 0);

if (closes.length < 20) throw new Error('Insufficient historical items');


// —— 下方完美對接你原有的計算邏輯 ——

const price = r.meta.regularMarketPrice || closes[closes.length - 1];

const high52 = Math.max(...closes);

const ddPct = ((price - high52) / high52) * 100;


const rsi14 = calcRSI(closes, 14);

const ma50 = closes.length >= 50 ? avg(closes.slice(-50)) : avg(closes);

const ma200 = closes.length >= 200 ? avg(closes.slice(-200)) : avg(closes);

const trendScore = calcTrend(price, ma50, ma200);


return { price, ddPct, rsi14, ma50, ma200, trendScore, dataLen: closes.length };


} catch (e) {

  console.error("Fetch implementation failed:", e);

  throw new Error(e.message);

}

}
```

```
// —— CURRENT RESULT STATE ——
```

```
let lastTicker='',lastBudget=0;
```

```
async function calculate(){
```

```
  // 💡 監控點 1：確認按鈕點擊有成功觸發函式
```

```
  alert("【Debug 1】成功進入 calculate 函式了！");
```

```
  const ticker = document.getElementById('ticker').value.trim().toUpperCase();
```

```
  const budget = parseFloat(document.getElementById('budget').value);
```

```
  const zh = lang === 'zh';
```

```
  const errBox = document.getElementById('errBox'), warnBox =  
document.getElementById('warnBox');
```

```
  errBox.classList.remove('show');
```

```
  warnBox.classList.remove('show');
```

```
  if(!ticker){
```

```
    errBox.textContent = zh ? '請輸入股票代號' : 'Please enter a ticker symbol';
```

```
    errBox.classList.add('show');
```

```
    return;
```

```
  }
```

```
  if(!budget || budget < 1){
```

```
    errBox.textContent = zh ? '請輸入每期預算' : 'Please enter your budget';
```

```
    errBox.classList.add('show');
```

```
return;  
}
```

```
const btn = document.getElementById('calcBtn');  
  
btn.disabled = true;  
  
btn.innerHTML = `<span class="spin"></span>${zh ? '載入市場數據...' : 'Loading  
market data...'}`;  
  
document.getElementById('result').classList.remove('show');
```

```
let rsi, ddPct, trendScore, price;
```

```
// 💡 監控點 2：確認準備去抓 API 之前的變數狀態
```

```
alert(`【Debug 2】輸入的代號是: ${ticker}，準備開始呼叫 API 函式...`);
```

```
try {
```

```
    // 呼叫我們上一輪改過名字的 API 函式
```

```
    const d = await getFreshTwelveDataPrice(ticker);
```

```
// 💡 監控點 3：確認有沒有成功從 API 函式拿到回傳值
```

```
alert("【Debug 3】成功拿到 API 數據了！最新價格是: " + d.price);
```

```
rsi = d.rsi14; ddPct = d.ddPct; trendScore = d.trendScore; price = d.price;
```

```
if(d.dataLen < 200){
```

```
    warnBox.textContent = zh ? `⚠️ ${ticker} 歷史數據較短（約  
${Math.round(d.dataLen/21)}個月）。` : `⚠️ ${ticker} has limited history`
```

```

(~${Math.round(d.dataLen/21)} months).`;

    warnBox.classList.add('show');

}

renderChips(ticker, d, zh);

} catch (e) {

    // 💡 監控點 4：捕捉抓資料過程中的任何噴錯

    alert("【Debug 錯誤攔截】在 fetch 過程中死掉了，原因: " + e.message);

    errBox.textContent = zh ? `無法取得 ${ticker} 數據：${e.message}` : `Could
not load ${ticker}: ${e.message}`;

    errBox.classList.add('show');

    btn.disabled = false;

    btn.innerHTML = `

```

```

    banner.className = 'rating-banner fu ' + tierClass[tier];

    document.getElementById('ratingDot').style.background = tierDotBg[tier];

    document.getElementById('ratingDot').textContent = T[lang].tierEmoji[tier];

    document.getElementById('ratingName').textContent = T[lang].tiers[tier];

    document.getElementById('ratingMult').textContent = `× ${baseMult.toFixed(1)}
    ${zh ? '建議倍數' : 'multiplier'}`;

    document.getElementById('rAmt').textContent = '$' + finalAmt.toLocaleString();

    document.getElementById('rSub').textContent = zh ? `基礎預算
    $$${budget.toLocaleString()} 美元` : `Base budget: $$${budget.toLocaleString()}`;

    const badge = document.getElementById('rBadge');

    if(baseMult > 1.05){ badge.className = 'r-badge up'; badge.textContent = `×
    ${finalMult}  ${zh ? '加碼' : 'Scale Up'}`; }

    else if(baseMult < 0.95){ badge.className = 'r-badge dn'; badge.textContent = `×
    ${finalMult}  ${zh ? '縮減' : 'Scale Down'}`; }

    else{ badge.className = 'r-badge nt'; badge.textContent = `× ${finalMult}  ${zh ?
    '標準' : 'Standard'}`; }

    const mi = document.getElementById('macroImpact');

    if(macroAdj > 0){ mi.className = 'macro-impact pos'; mi.textContent = zh ? `偏空
    判斷讓這期加了 +$$${Math.abs(Math.round(macroAdj))}` : `Bearish view added
    +$$${Math.abs(Math.round(macroAdj))} this period`; }

    else if(macroAdj < 0){ mi.className = 'macro-impact neg'; mi.textContent = zh ? `
    偏多判斷讓這期減了 -$$${Math.abs(Math.round(macroAdj))}` : `Bullish view
    reduced -$$${Math.abs(Math.round(macroAdj))} this period`; }

    else{ mi.className = 'macro-impact zero'; mi.textContent = zh ? '市場感受：中
    性，無調整' : 'Sentiment: Neutral — no adjustment'; }

```

```
setBar('sRSI', 'snRSI', 100 - rsi); setBar('sDD', 'snDD', Math.min(100, Math.max(0, -ddPct * 2.5 + 50))); setBar('sTrend', 'snTrend', trendScore);
```

```
document.getElementById('reasonList').innerHTML = '';
```

```
buildReasons(rsi, ddPct, trendScore, finalAmt, budget, macroAdj, zh).forEach(r => {
```

```
    const li = document.createElement('li'); li.innerHTML = `
```

```
    document.getElementById('reasonList').appendChild(li);
```

```
});
```

```
const period = document.getElementById('period').value, horizon =  
parseInt(document.getElementById('horizon').value), executed =  
parseInt(document.getElementById('executed').value);
```

```
simShares(budget, baseMult, Math.max(1, horizon - Math.min(executed, horizon -  
1)), price, zh);
```

```
updateSaveBtn(ticker);
```

```
document.getElementById('result').classList.add('show');
```

```
btn.disabled = false; btn.innerHTML = `
```

```
setTimeout(() => document.getElementById('result').scrollIntoView({ behavior:  
'smooth', block: 'start' }), 100);
```

```
}
```

```
function resetAdvisor(){  
  
    document.getElementById('result').classList.remove('show');  
  
    document.getElementById('ticker').value="";  
  
    document.getElementById('errBox').classList.remove('show');  
  
    document.getElementById('warnBox').classList.remove('show');  
  
    window.scrollTo({top:0,behavior:'smooth'});  
  
    document.getElementById('ticker').focus();  
  
}
```

```
function setBar(fid,nid,score){  
  
    const f=document.getElementById(fid);f.style.width=score+'%';  
  
    f.style.background=score>60?'#2d7a4f':score<40?'#b83232':'#c8813a';  
  
    document.getElementById(nid).textContent=Math.round(score);  
  
}
```

```
function renderChips(ticker,d,zh){  
  
    const sc=(id,val,sub,cls)=>{document.getElementById('c'+id).className='chip'+cls;document.getElementById('cv'+id).textContent=val;document.getElementById('cs'+id).textContent=sub;};  
  
    sc('Price','$'+d.price.toFixed(2),ticker,"");  
  
    sc('RSI',d.rsi14.toFixed(1),d.rsi14<30?(zh?'超賣':'Oversold'):d.rsi14>70?(zh?'超買':'Overbought'):(zh?'中性':'Neutral'),d.rsi14<30?'g':d.rsi14>70?'r':'o');  
  
    sc('DD',d.ddPct.toFixed(1)+'%',d.ddPct<-20?(zh?'大幅折價':'Deep dip'):d.ddPct<-8?(zh?'中度回撤':'Moderate'):(zh?'接近高點':'Near high'),d.ddPct<-20?'g':d.ddPct<-8?'o':'r');
```

```

    sc('Trend',d.price>d.ma50?(zh?'多頭':'Bull'):(zh?'空頭
': 'Bear'),d.trendScore>=60?'P>MA50&200':d.trendScore<=40?'P<MA200':'Mixed',d.tr
endScore>=60?'g':d.trendScore<=40?'r':'o');
}

```

```

function buildReasons(rsi,dd,trend,finalAmt,budget,macroAdj,zh){

```

```

    const r=[];

```

```

    if(rsi<25)r.push({icon:'🟢',text:zh?'RSI ${rsi.toFixed(0)}，極度超賣。市場正在恐
慌，對長期投資人來說往往是最好的買點。':'RSI ${rsi.toFixed(0)} — deeply
oversold. The market is fearful. Historically, fear creates the best long-term entry
points.``});

```

```

    else if(rsi<40)r.push({icon:'🟡',text:zh?'RSI ${rsi.toFixed(0)}，低於平均。賣壓偏
重，但股價已開始顯現吸引力。':'RSI ${rsi.toFixed(0)} — below average. Selling
pressure is elevated, but prices are starting to look attractive.``});

```

```

    else if(rsi<60)r.push({icon:'🟠',text:zh?'RSI ${rsi.toFixed(0)}，中性區間。沒有特
別訊號，維持正常投入步調。':'RSI ${rsi.toFixed(0)} — neutral territory. No
strong signal either way. Stay the course.``});

```

```

    else if(rsi<75)r.push({icon:'🔴',text:zh?'RSI ${rsi.toFixed(0)}，動能偏強。這個位
置的性價比稍低，保守一點。':'RSI ${rsi.toFixed(0)} — elevated momentum.
Risk/reward is slightly less favorable here.``});

```

```

    else r.push({icon:'🔴',text:zh?'RSI ${rsi.toFixed(0)}，超買。市場情緒過熱，這期
少投一點是對的。':'RSI ${rsi.toFixed(0)} — overbought. Market euphoria is high.
Discipline over excitement.``});

```

```

    if(dd<-25)r.push({icon:'💚',text:zh?'比 52 週高點便宜了
${Math.abs(dd).toFixed(1)}%。大幅折價，難得的長線機會。
':'${Math.abs(dd).toFixed(1)}% below its 52-week high — a significant discount.
These levels have historically rewarded patience.``});

```

```

    else if(dd<-8)r.push({icon:'🟡',text:zh?'距 52 週高點回撤
${Math.abs(dd).toFixed(1)}%，有一定折價空間。':'Down

```



```

    ${Math.abs(dd).toFixed(1)}% from its 52-week high — a moderate discount.`});

    else r.push({icon:'⚠️',text:zh?'距 52 週高點僅 ${Math.abs(dd).toFixed(1)}% ，接近高點。短期回撤機率偏高。':`Only ${Math.abs(dd).toFixed(1)}% off its 52-week high — near peak. Short-term pullbacks are statistically more likely.`});

    if(trend>=60)r.push({icon:'📈',text:zh?'股價站在 50 日及 200 日均線之上，長期趨勢健康。':`Price is above both the 50-day and 200-day moving averages. Long-term trend is intact.`});

    else if(trend<=40)r.push({icon:'📉',text:zh?'股價位於均線之下，下降趨勢中。已納入計算。':`Price is below key moving averages — downtrend in place. Factored into the calculation.`});

    else r.push({icon:'➡️',text:zh?'均線訊號混合，趨勢不明，使用中性值。':`Moving average signals are mixed — applying a neutral trend reading.`});

    if(macroAdj!==(0))r.push({icon:'🌐',text:zh?'你的市場感受（ ${macroAdj>0?'偏空':'偏多'} ）讓這期金額${macroAdj>0?'增加':'減少'}了${Math.abs(Math.round(macroAdj))}。':`Your ${macroAdj>0?'bearish':'bullish'} sentiment ${macroAdj>0?'added':'reduced'} ${Math.abs(Math.round(macroAdj))} this period.`});

    r.push({icon:'💡',text:zh?'綜合所有訊號，這期建議投入${finalAmt.toLocaleString()} 美元。':`Bottom line: invest ${finalAmt.toLocaleString()} this period based on current conditions.`});

    return r;
}

```

```
let chartInst=null;
```

```
function simShares(budget,mult,periods,price,zh){
```

```
    const labels=[],fixedS=[],smartS=[];let fS=0,sS=0;
```

```
    const step=Math.max(1,Math.floor(periods/24)),seed=price*0.015;let simP=price;
```

```
    for(let i=1;i<=periods;i++){
```

```

simP=Math.max(price*.6,simP+(Math.sin(i*.7)*.5+Math.cos(i*.4)*.3)*seed);

const pm=Math.max(.4,Math.min(1.95,mult+(Math.sin(i*1.2)*.12)));

fS+=budget/simP;sS+=(budget*pm)/simP;

if(i%step===0||i===periods){labels.push(`P${i}`);fixedS.push(parseFloat(fS.toFixed(3))
);smartS.push(parseFloat(sS.toFixed(3)));}

}

const ff=fixedS[fixedS.length-1],sf=smartS[smartS.length-1];

document.getElementById('cFixed').textContent=ff.toFixed(2)+' sh';

document.getElementById('cSmart').textContent=sf.toFixed(2)+' sh';

document.getElementById('cDiff').textContent=`+${((sf-ff)/ff*100).toFixed(1)}%
${zh?'更多股數':'more shares'}`;

if(chartInst){chartInst.destroy();chartInst=null;}

const ctx=document.getElementById('growthChart').getContext('2d');

chartInst=new Chart(ctx,{type:'line',data:{labels,datasets:[{label:zh?'傳統定投
':'Fixed
DCA',data:fixedS,borderColor:'#aba69e',backgroundColor:'rgba(171,166,158,0.08)',fil
l:true,tension:.4,pointRadius:0,borderWidth:2},{label:zh?'智能定投':'Smart
DCA',data:smartS,borderColor:'#2d7a4f',backgroundColor:'rgba(45,122,79,0.09)',fill:t
rue,tension:.4,pointRadius:0,borderWidth:2.5}]},options:{responsive:true,interaction
:{mode:'index',intersect:false},plugins:{legend:{labels:{color:'#6b6560',font:{size:11},
boxWidth:12}},tooltip:{backgroundColor:'#fff',borderColor:'#e0dbd0',borderWidth:1,
titleColor:'#6b6560',bodyColor:'#1a1814',callbacks:{label:c=>` ${c.dataset.label}:
${c.raw.toFixed(2)}
sh`}}},scales:{x:{ticks:{color:'#aba69e',font:{size:10},maxTicksLimit:6},grid:{color:'rgba
(0,0,0,0.04)'}},y:{ticks:{color:'#aba69e',font:{size:10},callback:v=>v.toFixed(1)+'
sh'},grid:{color:'rgba(0,0,0,0.05)'}}}}});

}

```

```
// —— WATCHLIST (localStorage) ——
```

```
function getWatchlist(){try{return  
JSON.parse(localStorage.getItem('pi_watchlist') || '[]');}catch{return[];}}
```

```
function saveWatchlist(wl){localStorage.setItem('pi_watchlist',JSON.stringify(wl));}
```

```
function toggleSave(){  
  
    const ticker=lastTicker;const budget=lastBudget;  
  
    if(!ticker)return;  
  
    let wl=getWatchlist();  
  
    const idx=wl.findIndex(w=>w.ticker===ticker);  
  
    if(idx>=0){wl.splice(idx,1);}  
  
    else{wl.push({ticker,budget});}  
  
    saveWatchlist(wl);updateSaveBtn(ticker);  
  
}
```

```
function updateSaveBtn(ticker){  
  
    const wl=getWatchlist();  
  
    const isSaved=wl.some(w=>w.ticker===ticker);  
  
    const btn=document.getElementById('saveBtn');  
  
    const zh=lang==='zh';  
  
    if(isSaved){btn.className='act-btn saved';btn.innerHTML=`  
<span>${T[lang].saved}</span>`;}  
  
    else{btn.className='act-btn';btn.innerHTML=` <span id="t-  
save">${T[lang].save}</span>`;}
```

```
}
```

```
async function renderWatchlist(){
```

```
    const wl=getWatchlist();
```

```
    const zh=lang==='zh';
```

```
    const cont=document.getElementById('wlContent');
```

```
    if(wl.length===0){
```

```
        cont.innerHTML=`<div class="wl-empty">${zh?'尚未加入任何標的。<br>先分析一支股票，再點「加入自選清單」。':'No stocks saved yet.<br>Analyze a stock first, then tap "Save to Watchlist".'}</div>`;return;
```

```
    }
```

```
}
```

```
cont.innerHTML=wl.map(w=>`
```

```
    <div class="wl-card" id="wlcards-${w.ticker}">
```

```
        <div class="wl-ticker">${w.ticker}</div>
```

```
        <div class="wl-info">
```

```
            <div class="wl-budget">${zh?'預算':'Budget'}:
```

```
            ${w.budget.toLocaleString()}</div>
```

```
            <div class="wl-amount" id="wlamts-${w.ticker}">—</div>
```

```
            <div class="wl-badge" id="wlbads-${w.ticker}"></div>
```

```
            <div class="wl-loading" id="wlload-${w.ticker}">⌚ ${zh?'載入中...'
```

```
            ':'Loading...'}</div>
```

```
        </div>
```

```
        <button class="wl-remove" onclick="removeFromWatchlist('${w.ticker}')"`
```

```
        title="Remove">✕</button>
```

```

    </div>`).join("");

    // Load each
    for(const w of wl){await loadWatchlistItem(w);}
}

async function loadWatchlistItem(w){
    const zh=lang==='zh';

    try{
        const d=await getFreshTwelveDataPrice(w.ticker);
        const mult=calcMult(d.rsi14,d.ddPct,d.trendScore);
        const amt=Math.round(w.budget*mult/10)*10;
        const tier=getTier(mult);
        const amtEl=document.getElementById('wlamt-'+w.ticker);
        const badgeEl=document.getElementById('wlbade-'+w.ticker);
        const loadEl=document.getElementById('wlload-'+w.ticker);

        if(amtEl){amtEl.textContent='$'+amt.toLocaleString();amtEl.style.color=tierDotBg[tier];}

        if(badgeEl){
            badgeEl.textContent=`${T[lang].tierEmoji[tier]} ${T[lang].tiers[tier]} · ×
            ${mult.toFixed(1)}`;

            badgeEl.style.background=tierDotBg[tier]+'22';
            badgeEl.style.color=tierDotBg[tier];
            badgeEl.style.border='1px solid '+tierDotBg[tier]+'66';
            badgeEl.style.borderRadius='20px';badgeEl.style.padding='3px

```

```

10px';badgeEl.style.fontSize='11px';badgeEl.style.fontWeight='600';

    }

    if(loadEl)loadEl.style.display='none';

}catch(e){

    const loadEl=document.getElementById('wload-'+w.ticker);

    if(loadEl){loadEl.textContent=zh?'載入失敗':'Failed to
load';loadEl.style.color='#b83232';}

}

}

function removeFromWatchlist(ticker){

    let
    wl=getWatchlist();wl=wl.filter(w=>w.ticker!==ticker);saveWatchlist(wl);renderWatchli
st();

}

async function refreshWatchlist(){

    const btn=document.getElementById('refreshBtn');

    const zh=lang==='zh';

    btn.disabled=true;btn.innerHTML=`<span class="spin spin-dk"></span>${zh?'更新
中...':'Refreshing...'}`;

    await renderWatchlist();

    btn.disabled=false;btn.innerHTML=` <span id="t-
refresh">${T[lang].refresh}</span>`;

}

```

```
// —— COPY ——
```

```
function copyResult(){

    const zh=lang==='zh';

    const ticker=lastTicker||'—';

    const amt=document.getElementById('rAmt').textContent;

    const badge=document.getElementById('rBadge').textContent;

    const rating=document.getElementById('ratingName').textContent;

    const mi=document.getElementById('macroImpact').textContent;

    const reasons=[...document.querySelectorAll('#reasonList
li')].map(li=>li.textContent.trim()).join('\n');

    const text=zh?`【PulseInvest 智能定投建議】\n${ticker} — ${rating}\n 本期建議
投入：${amt}  ${badge}\n${mi}\n\n 建議理由：\n${reasons}\n\n ⚠️ 非投資建
議，過去不代表未來。`:[PulseInvest Smart DCA]\n${ticker} —
${rating}\nSuggested: ${amt}  ${badge}\n${mi}\n\nWhy:\n${reasons}\n\n ⚠️ Not
investment advice. Past ≠ future results.`;

    navigator.clipboard.writeText(text).then(()=>{document.getElementById('copyBtn').te
xtContent='✓
Copied!';setTimeout(()=>{document.getElementById('copyBtn').innerHTML=`<span
id="t-cpbtn">${T[lang].cpbtn}</span>`;},2000);}).catch(()=>alert(text));

}
```

```
// —— BACKTEST ——
```

```
let btChartInst=null;

async function runBacktest(){

    const ticker=document.getElementById('btTicker').value.trim().toUpperCase();
```

```
const startStr=document.getElementById('btStart').value.trim();

const endStr=document.getElementById('btEnd').value.trim();

const budget=parseFloat(document.getElementById('btBudget').value);

const zh=lang==='zh';

if(!ticker||!startStr||!endStr||!budget){alert(zh?'請填寫所有欄位':'Please fill in
all fields');return;}

const btn=document.getElementById('btBtn');

btn.disabled=true;btn.innerHTML=`<span class="spin spin-dk"></span>${zh?'載入
中...':'Loading...'}`;

try{

    const[sy,sm]=startStr.split('-').map(Number);const[ey,em]=endStr.split('-
').map(Number);

    if(!sy||!sm||!ey||!em)throw new Error(zh?'日期格式錯誤':'Invalid date
format');

    const p1=Math.floor(new Date(sy,sm-1,1).getTime()/1000),p2=Math.floor(new
Date(ey,em,1).getTime()/1000);

    const btUrls=[

`https://query1.finance.yahoo.com/v8/finance/chart/${ticker}?interval=1mo&period
1=${p1}&period2=${p2}`,

`https://query2.finance.yahoo.com/v8/finance/chart/${ticker}?interval=1mo&period
1=${p1}&period2=${p2}`

    ];

    let btJson=null;

    for(const u of btUrls){try{const r2=await
fetch(u,{headers:{'Accept':'application/json'}});if(!r2.ok)continue;const j2=await
```



```
r2.json());if(j2.chart&&j2.chart.result&&j2.chart.result[0]){btJson=j2;break;}}catch(e){
continue;}}
```

```
if(!btJson)throw new Error(zh?'無法取得歷史數據':'Could not load historical
data');
```

```
const btR=btJson.chart.result[0];
```

```
const rawC=btR.indicators.quote[0].close;
```

```
const rawT=btR.timestamp;
```

```
const closes=[],timestamps=[];
```

```
rawC.forEach((v,i)=>{if(v!==null&&v>0){closes.push(v);timestamps.push(rawT[i]);}});
```

```
if(closes.length<3)throw new Error(zh?'找不到 ${ticker} 的歷史數據':'No
historical data for ${ticker}');
```

```
const btL=[],btF=[],btS=[];let fSh=0,sSh=0,fSp=0,sSp=0;
```

```
for(let i=0;i<closes.length;i++){
```

```
const price=closes[i];if(!price||price<=0)continue;
```

```
const av=closes.slice(0,i+1).filter(v=>v&&v>0);
```

```
const ri=av.length>=15?calcRSI(av,Math.min(14,av.length-1)):50;
```

```
const hi=Math.max(...av.slice(-Math.min(12,av.length)));
```

```
const dd=(price-hi)/hi*100;
```

```
const m50=av.length>=50?avg(av.slice(-
50)):avg(av),m200=av.length>=200?avg(av.slice(-200)):avg(av);
```

```
const ts=calcTrend(price,m50,m200),mult=calcMult(ri,dd,ts);
```

```
fSh+=budget/price;fSp+=budget;sSh+=(budget*mult)/price;sSp+=budget*mult;
```

```
const d=new Date(timestamps[i]*1000);
```

```
btL.push(`${d.getFullYear()}-${String(d.getMonth()+1).padStart(2,'0')}`);
```

```

    btF.push(parseFloat(fSh.toFixed(3)));btS.push(parseFloat(sSh.toFixed(3)));

}

const lp=closes.filter(v=>v&&v>0).slice(-1)[0];

const fV=fSh*lp,sV=sSh*lp;

const ac1=(fSp/fSh).toFixed(2),ac2=(sSp/sSh).toFixed(2);

const fmt=n=>'${Math.round(n).toLocaleString()};

const rows=zh?['傳統定投：總投入',fmt(fSp),'],['智能定投：總投入',fmt(sSp),'],['傳統定投：累積股數',fSh.toFixed(3)+' sh','],['智能定投：累積股數',sSh.toFixed(3)+' sh','g'],['傳統定投：平均成本','$+ac1,'],['智能定投：平均成本','$+ac2,parseFloat(ac2)<parseFloat(ac1)?'g':'],['傳統定投：期末估值',fmt(fV),'],['智能定投：期末估值',fmt(sV),'g'],['多累積股數','+'+(((sSh-fSh)/fSh)*100).toFixed(1)+'%', 'g'],['期末估值差距','+'+(((sV-fV)/fV)*100).toFixed(1)+'%', 'g']]:[['Fixed DCA: Total Invested',fmt(fSp),'],['Smart DCA: Total Invested',fmt(sSp),'],['Fixed DCA: Shares',fSh.toFixed(3)+' sh','],['Smart DCA: Shares',sSh.toFixed(3)+' sh','g'],['Fixed DCA: Avg Cost','$+ac1,'],['Smart DCA: Avg Cost','$+ac2,parseFloat(ac2)<parseFloat(ac1)?'g':'],['Fixed DCA: Final Value',fmt(fV),'],['Smart DCA: Final Value',fmt(sV),'g'],['Extra Shares','+'+(((sSh-fSh)/fSh)*100).toFixed(1)+'%', 'g'],['Value Difference','+'+(((sV-fV)/fV)*100).toFixed(1)+'%', 'g']];

document.getElementById('btRows').innerHTML=rows.map(([k,v,cl])=><div class="bt-row"><span class="bt-key">${k}</span><span class="bt-val ${cl}">${v}</span></div>`).join('');

document.getElementById('btResult').style.display='block';

if(btChartInst){btChartInst.destroy();btChartInst=null;}

const ctx2=document.getElementById('btChart').getContext('2d');

btChartInst=new Chart(ctx2,{type:'line',data:{labels:btL,datasets:[{label:zh?'傳統定投':'Fixed DCA',data:btF,borderColor:'#aba69e',backgroundColor:'rgba(171,166,158,0.08)',fill:true,tension:.3,pointRadius:0,borderWidth:2},{label:zh?'智能定投':'Smart

```

```
DCA',data:btS,borderColor:'#2d7a4f',backgroundColor:'rgba(45,122,79,0.1)',fill:true,t  
ension:.3,pointRadius:0,borderWidth:2.5]],options:{responsive:true,interaction:{mo  
de:'index',intersect:false},plugins:{legend:{labels:{color:'#6b6560',font:{size:11},box  
Width:12}},tooltip:{backgroundColor:'#fff',borderColor:'#e0dbd0',borderWidth:1,title  
Color:'#6b6560',bodyColor:'#1a1814',callbacks:{label:c=>` ${c.dataset.label}:  
${c.raw.toFixed(2)}  
sh`}},scales:{x:{ticks:{color:'#aba69e',font:{size:10},maxTicksLimit:8},grid:{color:'rgba  
(0,0,0,0.04)'}},y:{ticks:{color:'#aba69e',font:{size:10},callback:v=>v.toFixed(1)+'  
sh'},grid:{color:'rgba(0,0,0,0.05)'}}}}}});
```

```
setTimeout(()=>document.getElementById('btResult').scrollIntoView({behavior:'smo  
oth',block:'start'}),100);
```

```
    }catch(e){alert(zh?'回測失敗 : ${e.message}':'Backtest failed: ${e.message}');}
```

```
    btn.disabled=false;btn.innerHTML=`<span id="t-btbtn">${T[lang].btbtn}</span>`;
```

```
  }
```

```
</script>
```

```
</body>
```

```
</html>
```